

AI STUDYBUDDY

AI-Powered Personalized Learning Platform

Project Documentation

1. Introduction

AI StudyBuddy is an AI-powered learning assistance platform designed to simplify the study process. The reference project uses a React frontend, Node.js and Express.js backend, MongoDB with Mongoose, JWT authentication, bcrypt-based password protection, and Google Gemini AI for educational content generation. The reference system supports summaries, flashcards, quizzes and personalized study plans.

The proposed workflow with a Study Profile, material-processing pipeline, semantic retrieval, performance analysis, recommendations, weekly progress, and points/streaks. These additional workflow elements are treated as proposed project features, Project Architecture, AI integration, database entities, API endpoints, and execution flow. ☞

2. Problem Statement

Students often work with lecture notes, PDFs, textbooks and other learning resources. The reference case study identifies difficulty understanding lengthy materials, manual note preparation, repetitive flashcard/quiz creation, lack of personalized study plans, and the need to switch between multiple applications.

- Difficulty understanding lengthy study materials within limited time.
- Manual preparation of notes consumes study time.
- Creating flashcards and revision questions is repetitive.
- Personalized study planning is difficult to maintain manually.
- Students need a centralized place for AI-generated learning resources.
- Students need a simple way to assess their understanding.

These points are directly aligned with the problem section.

3. Objectives

- Provide secure student registration and login.
- Maintain a student Study Profile containing learning information.
- Allow study materials to be uploaded and processed.
- Use AI to generate summaries, examples, flashcards, quizzes and study plans.
- Measure quiz performance and identify strong and weak topics.
- Generate personalized recommendations from performance data.
- Track weekly progress, points and study streaks as proposed engagement features.

4. PROJECT ARCHITECTURE

AI StudyBuddy - Proposed System Architecture

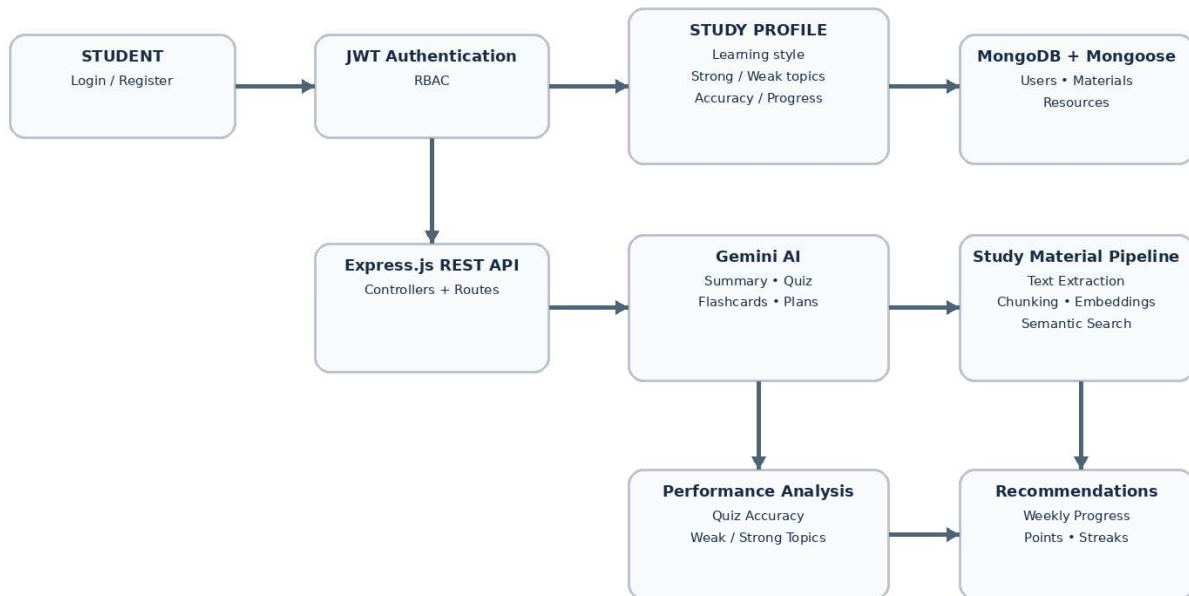


Figure 1. Proposed system architecture based on the reference backend and the supplied workflow.

5. Major Modules

5.1 Student Authentication

Students register and log in before accessing protected learning features. JWT is used for authentication and RBAC separates student and administrator privileges.

5.2 Study Profile

The proposed profile stores learning style, strong topics, weak topics, quiz accuracy, study progress and learning history. It becomes the personalization layer for recommendations.

5.3 Study Material Management

Students upload or enter study materials. The reference project stores uploaded materials in MongoDB and exposes REST APIs for material management.

5.4 AI Chatbot

The chatbot uses retrieved study content as context and communicates with Gemini to provide learning assistance.

5.5 Learning Content Generation

The reference system supports AI-generated summaries, flashcards, quizzes and study plans. The proposed workflow adds story-style explanations, examples, concept maps and recaps as learning-output formats.

5.6 Quiz & Assessment

AI-generated multiple-choice questions are used to assess understanding. Quiz results are then used by the proposed analytics layer.

5.7 Performance Analysis

The system compares quiz performance across topics and classifies topics as strong or weak using defined project thresholds.

5.8 Recommendation Engine

Weak topics can trigger revision suggestions, practice questions, recap sessions or a study plan.

5.9 Progress & Gamification

Weekly progress, points and streaks are proposed features for encouraging consistent study activity.

5.10 Settings and Light/Dark Mode

The Settings module allows students to control application preferences.

Settings

- Account settings
- Profile settings
- Notification preferences
- Privacy preferences
- Focus-mode preferences
- AI preferences
- Security settings
- Theme settings

6. AI Material Processing Pipeline

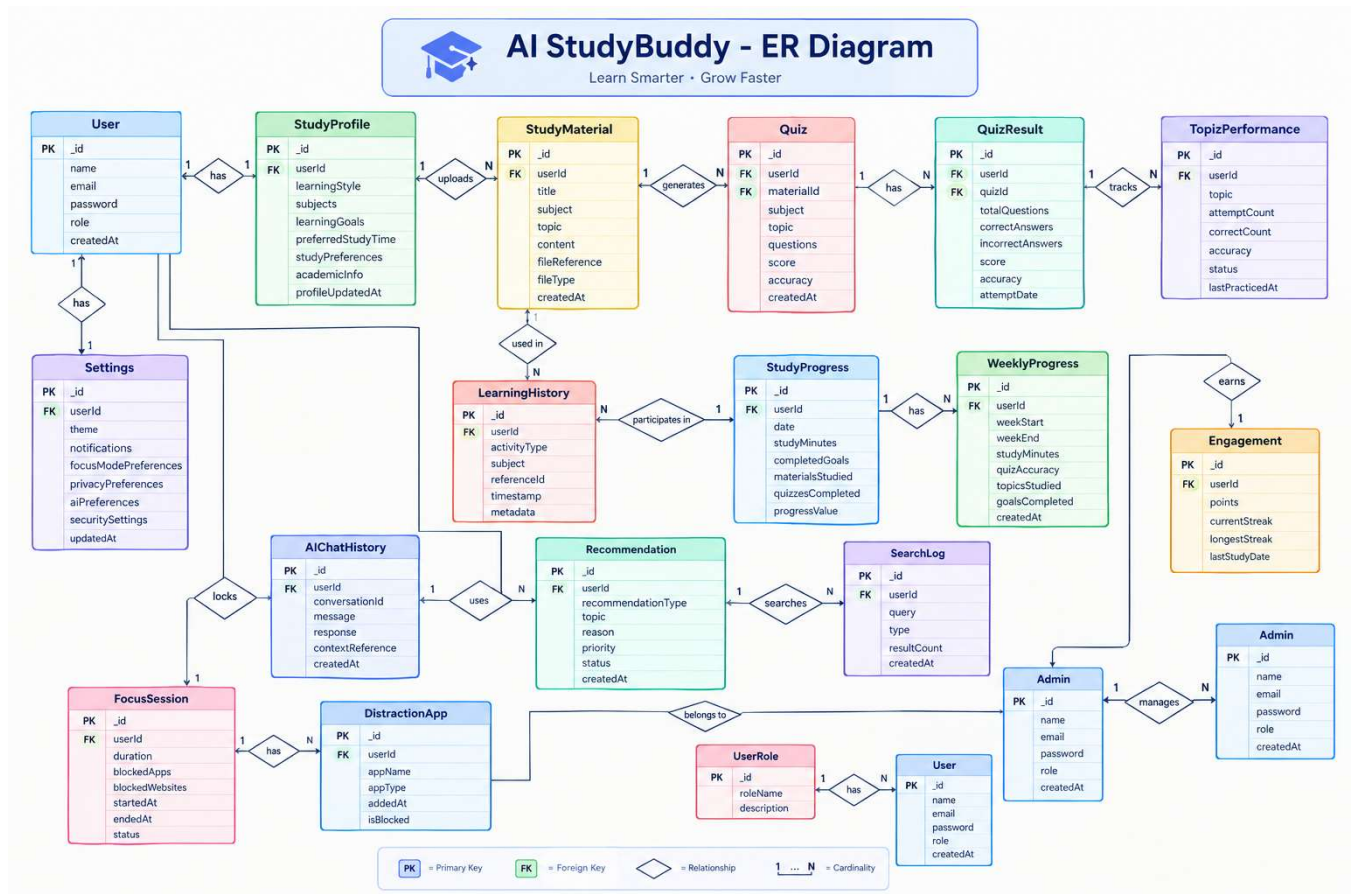
AI Learning Pipeline - Example



Figure 2. Example pipeline for turning study material into personalized learning support.

Important implementation note: the reference document explicitly describes file upload and Gemini generation, but it does not explicitly document embeddings or semantic search. Those two stages are therefore proposed additions in this workflow.

6.1 ENTITY RELATIONSHIP (ER) DIAGRAM



Database Entities

1 User

Attributes:

- _id
- name
- email
- password
- role
- createdAt

2 Study Profile

Attributes:

- `_id`
- `userId`
- `learningStyle`
- `learningGoals`
- `subjects`
- `preferredStudyTime`
- `profileUpdatedAt`

3 Study Material

Attributes:

- `_id`
- `userId`
- `title`
- `subject`
- `content`
- `fileReference`
- `createdAt`

4 Quiz

Attributes:

- `_id`
- `userId`
- `materialId`
- `subject`
- `topic`
- `questions`
- `score`
- `accuracy`
- `createdAt`

5 Topic Performance

Attributes:

- `_id`
 - `userId`
 - `topic`
 - `attemptCount`
 - `correctCount`
 - `accuracy`
 - `status`
 - `lastPracticedAt`
-

6 Study Progress

Attributes:

- `_id`
 - `userId`
 - `date`
 - `studyMinutes`
 - `completedGoals`
 - `materialsStudied`
 - `quizzesCompleted`
 - `progressValue`
-

7 Learning History

Attributes:

- `_id`
 - `userId`
 - `activityType`
 - `subject`
 - `referenceId`
 - `timestamp`
 - `metadata`
-

8 AI Chat History

Attributes:

- `_id`
- `userId`
- `conversationId`
- `message`
- `response`

- contextReference
- createdAt

7. Functional Requirements

ID	Requirement	Description
FR-01	User Registration	The system shall allow a student to register using required account information.
FR-02	User Login	The system shall authenticate users and issue a JWT access token.
FR-03	Study Material Upload	Authenticated students shall be able to upload study materials.
FR-04	AI Summary	The system shall generate concise summaries from study content.
FR-05	Flashcards	The system shall generate question-answer flashcards.
FR-06	Quiz	The system shall generate multiple-choice questions from study content.
FR-07	Study Plan	The system shall generate a study plan based on goals and available study time.
FR-08	Study Profile	The proposed system shall maintain learning-style, topic-performance and progress information.
FR-09	Performance Analysis	The proposed system shall calculate quiz performance and identify topic-level strengths and weaknesses.
FR-10	Recommendations	The proposed system shall recommend next study actions using profile and performance data.

8. Non-Functional Requirements

Security

Student data must be protected from unauthorized access.

Performance

The system should provide responsive API and dashboard operations.

Scalability

The modular architecture should allow additional AI and learning features to be added.

Usability

The interface should be simple and easy for students to understand.

Maintainability

Frontend, backend, AI, database, search, and analytics components should remain modular.

Reliability

Student learning data should be stored reliably and remain available after successful operations.

9. Example Study Profile

Example Study Profile

Rahul - B.Sc. Computer Science	
Learning Style	Visual + Practice-based
Strong Topics	Python, DBMS
Weak Topics	Statistics, Data Structures
Quiz Accuracy	82%
Study Progress	68%
Learning History	12 quizzes • 8 summaries • 5 study plans

Figure 3. Example profile screen. Values are illustrative.

10. Learning Content Generated by AI

Output	Purpose
Story	Explain a difficult concept through a short, student-friendly story.
Examples	Give simple examples, code snippets or real-world applications where appropriate.
Concept Map	Show relationships between major and minor concepts.
Recap	Provide a short revision section containing the most important points.
Quiz	Generate objective questions for self-assessment.

Example Concept Map - Data Structures

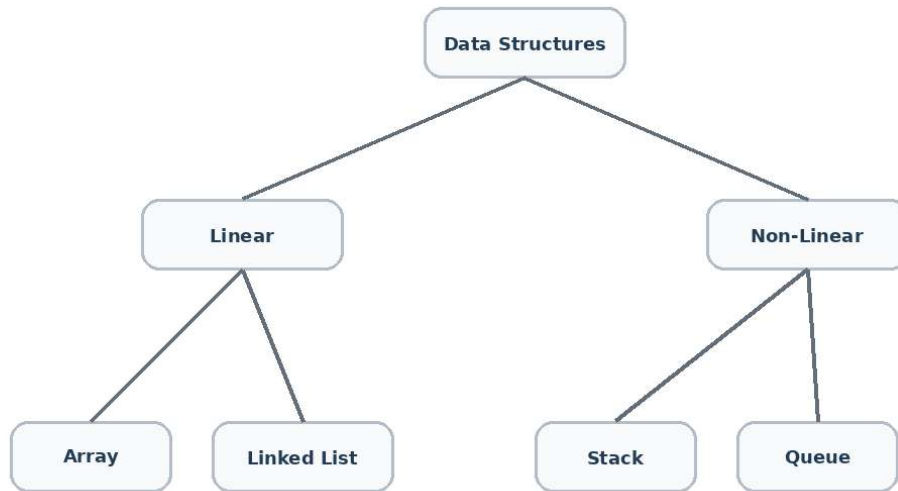
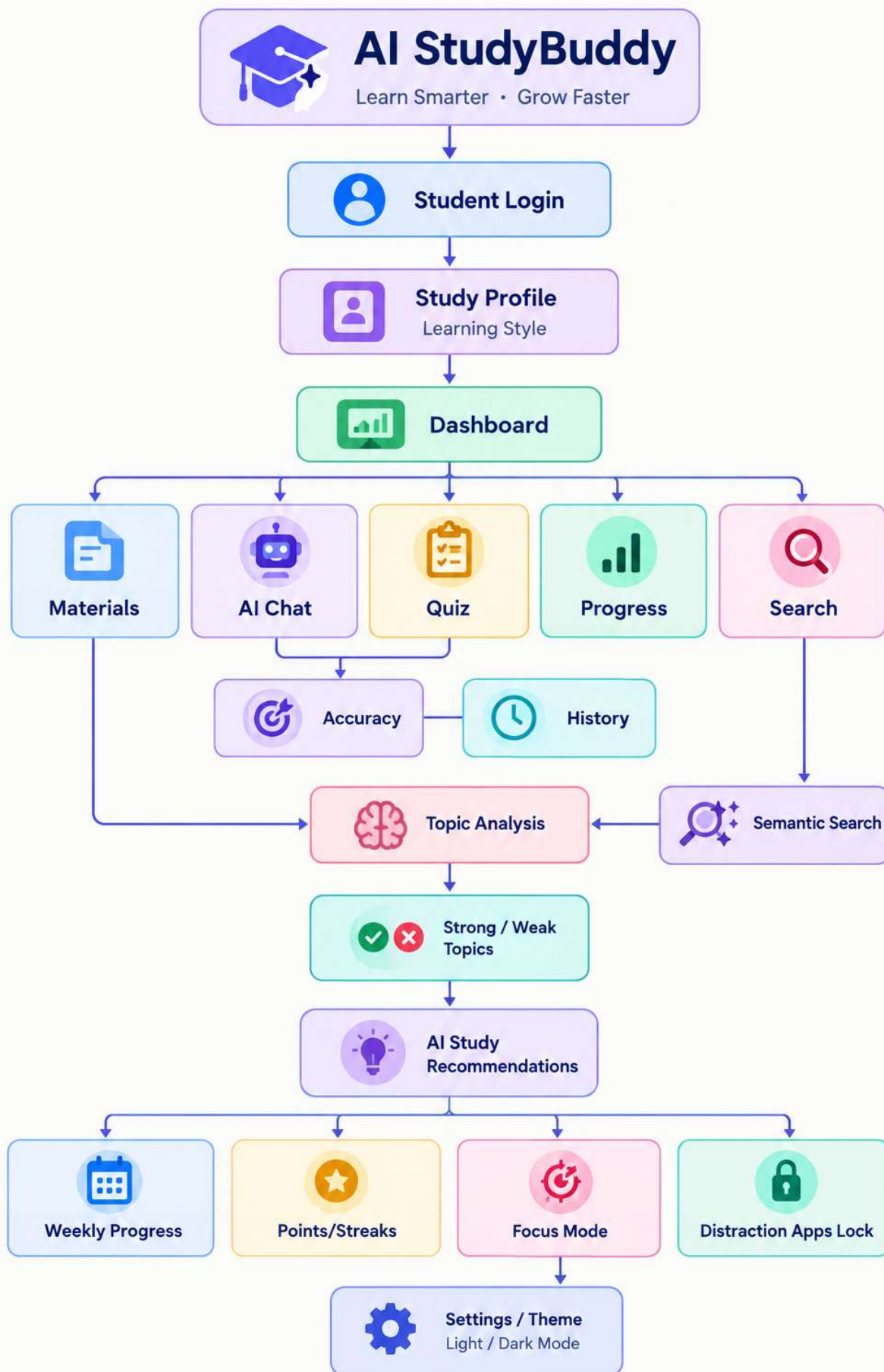


Figure 4. Example concept map generated as a learning aid.

USER FLOW DIAGRAM



11. Quiz and Performance Analysis

Example AI Quiz Screen



The image shows a quiz interface with a light blue background. At the top, it says "Question 4 of 10". Below that is the question: "Which data structure follows the FIFO principle?". There are four options, each in a rounded rectangular button: "A. Stack", "B. Queue", "C. Tree", and "D. Graph". At the bottom, there is a small text line: "After submission: score → topic analysis → recommendation".

Question 4 of 10

Which data structure follows the FIFO principle?

A. Stack

B. Queue

C. Tree

D. Graph

After submission: score → topic analysis → recommendation

Figure 5. Example quiz interface. The question and options are illustrative.

After submission, the system can record the student's score and map each question to a topic. For example, if a student answers 8 out of 10 questions correctly, quiz accuracy is 80%. The project can then aggregate accuracy by topic and update the Study Profile.

Example rule (proposed): Strong topic = accuracy $\geq 75\%$; Weak topic = accuracy $< 60\%$; scores in between can be marked as 'Needs Practice'. The actual thresholds should be finalized during implementation.

12 Recommendation Flow

1. Quiz is completed.
2. Answers and score are stored.
3. Questions are mapped to topics.
4. Topic-wise accuracy is calculated.
5. Weak topics are identified.
6. The recommendation engine selects suitable actions.
7. The Study Profile is updated.
8. The next weekly progress view includes the updated activity.

13. Example Dashboard

Example Dashboard - Student View

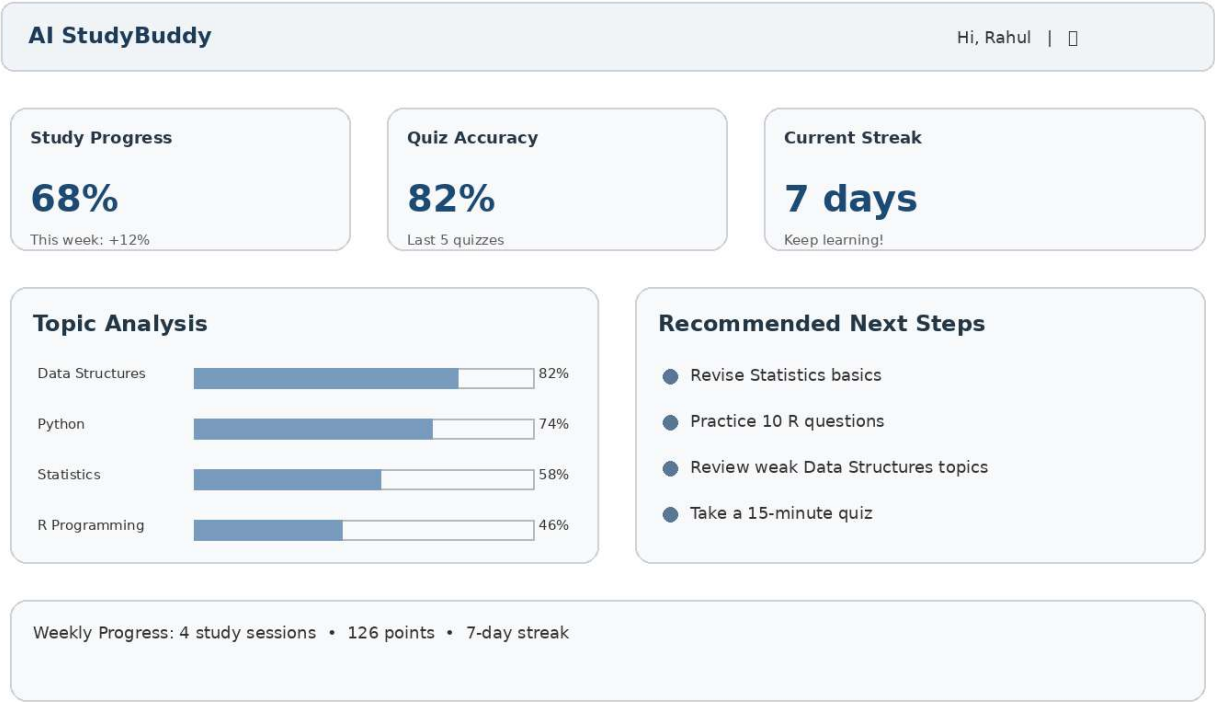


Figure 6. Example dashboard showing progress, quiz accuracy, streaks and recommendations. Values are illustrative.

14. Database Design

The project defines User, Study Material, Summary, Flashcard, Quiz and Study Plan entities. User-to-material and user-to-resource relationships support retrieval of a student's generated resources.

Entity	Example Fields	Purpose
User	<code>_id</code> , name, email, password, role, <code>createdAt</code>	Account and authorization information
StudyMaterial	<code>_id</code> , <code>userId</code> , title, subject, content, <code>createdAt</code>	Uploaded learning content
Summary	<code>_id</code> , <code>userId</code> , <code>materialId</code> , summary, <code>createdAt</code>	AI-generated summary
Flashcard	<code>_id</code> , <code>userId</code> , <code>materialId</code> , question, answer, <code>createdAt</code>	AI-generated flashcards
Quiz	<code>_id</code> , <code>userId</code> , <code>materialId</code> , questions, <code>createdAt</code>	AI-generated quiz

	createdAt	
StudyPlan	_id, userId, studyPlan, examDate, createdAt	Personalized study schedule
StudyProfile*	userId, learningStyle, strongTopics, weakTopics, accuracy, progress	Proposed personalization data
Performance*	userId, quizId, topic, score, attemptedAt	Proposed analytics data
Recommendation*	userId, topic, action, createdAt	Proposed recommendation data

* Proposed entities added to support the supplied Study Profile and performance workflow.

?

15. Technology Stack

Technology	Use
Frontend	React.js
Backend	Node.js + Express.js
Database	MongoDB + Mongoose
AI	Google Gemini API
Authentication	JWT + bcryptjs
File Upload	Multer
API Testing	Postman / Thunder Client
Development	Visual Studio Code
Configuration	dotenv

16. API Design –Endpoints

Endpoint	Purpose
POST /api/auth/register	Register a user
POST /api/auth/login	Authenticate user and generate JWT
POST /api/material/upload	Upload study material
POST /api/materials/:id/summarize	Generate summary
POST /api/ai/flashcards	Generate flashcards

POST /api/ai/quiz

Generate quiz

POST /api/ai/study-plan

Generate study plan

?

17. Security

- JWT authentication protects private routes.
- Role-Based Access Control separates student and administrator permissions.
- Passwords are hashed/encrypted using bcryptjs before storage.
- Environment variables are used for database, JWT and Gemini configuration.
- Uploaded materials should be validated before processing.
- AI requests should be made through the backend rather than exposing secret API keys in the frontend.

18. Example Student Scenario

Rahul uploads a Data Structures PDF. The system extracts the material, identifies relevant sections, and sends the required context to the AI service. The student receives a concise explanation, examples, a concept map, recap notes and a quiz. Rahul completes the quiz with 6/10 correct. The system maps the incorrect questions to topics and updates his profile. If Queue and Linked List are weak topics, the next recommendation can focus on those topics with a short recap followed by another practice quiz. After the study session, weekly progress and streak information are updated.

The student scenario and the system's upload → AI generation → saved resources flow are grounded. The profile analysis, semantic search and gamification steps in this example are proposed extensions.

19. Testing Plan

Test Area	Example Tests
Authentication	Register, login, invalid password, expired token, role restrictions
Upload	Valid PDF/text, unsupported file, empty file, large file
AI Generation	Summary, flashcards, quiz and study plan generation
Quiz	Answer submission, score calculation, topic mapping
Profile	Accuracy/progress update after quiz
Recommendation	Weak-topic detection and recommendation generation
Security	Unauthorized API access, protected routes, secret handling
Database	Create, read, update and retrieve user resources

20. Expected Outcome

The completed system is intended to provide a centralized learning environment in which students can upload materials, generate AI-assisted learning resources, practice through quizzes, and review their progress.

With the proposed Study Profile and performance loop, the project can be extended from a content-generation tool into a more personalized learning workflow: learn → practice → analyze → recommend → update profile → track progress.

21. Conclusion

AI StudyBuddy combines secure web application architecture with Generative AI to support student learning. The implementation provides the core foundation—authentication, study-material management, MongoDB persistence, Gemini integration, summaries, flashcards, quizzes and study plans. The supplied workflow adds a personalization loop through Study Profile, semantic retrieval, performance analysis, recommendations, progress tracking and gamification. Together, these components form a clear end-to-end model for an academic AI learning assistant.